



MATH230: Applications of Double Integrals

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Area and volume

1. Find the area of the rectangle $R = [0, 5] \times [0, 4]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [0, 5] \times [0, 4]$$

Answer: _____

2. Find the volume of the solid under $z = 8$ and above the rectangle $[0, 3] \times [1, 3]$ in the xy -plane.

$$\iint_R 8 \, dA, [0, 3] \times [1, 3]$$

Answer: _____

3. Find the area of the rectangle $R = [1, 6] \times [1, 3]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 6] \times [1, 3]$$

Answer: _____

4. Find the volume of the solid under $z = 7$ and above the rectangle $[0, 5] \times [1, 4]$ in the xy -plane.

$$\iint_R 7 \, dA, [0, 5] \times [1, 4]$$

Answer: _____

5. Find the area of the rectangle $R = [0, 7] \times [2, 7]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [0, 7] \times [2, 7]$$

Answer: _____



6. Find the volume of the solid under $z = 9$ and above the rectangle $[1, 4] \times [0, 5]$ in the xy -plane.

$$\iint_R 9 \, dA, [1, 4] \times [0, 5]$$

Answer: _____

7. Find the area of the rectangle $R = [2, 7] \times [0, 4]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [2, 7] \times [0, 4]$$

Answer: _____

8. Find the volume of the solid under $z = 5$ and above the rectangle $[1, 5] \times [1, 5]$ in the xy -plane.

$$\iint_R 5 \, dA, [1, 5] \times [1, 5]$$

Answer: _____

9. Find the area of the rectangle $R = [1, 7] \times [1, 6]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 7] \times [1, 6]$$

Answer: _____

10. Find the volume of the solid under $z = 2$ and above the rectangle $[1, 4] \times [1, 3]$ in the xy -plane.

$$\iint_R 2 \, dA, [1, 4] \times [1, 3]$$

Answer: _____

11. Find the area of the rectangle $R = [1, 7] \times [2, 3]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 7] \times [2, 3]$$

Answer: _____

12. Find the volume of the solid under $z = 5$ and above the rectangle $[0, 5] \times [0, 3]$ in the xy -plane.

$$\iint_R 5 \, dA, [0, 5] \times [0, 3]$$

Answer: _____

13. Find the area of the rectangle $R = [1, 7] \times [0, 5]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 7] \times [0, 5]$$

Answer: _____



14. Find the volume of the solid under $z = 4$ and above the rectangle $[0,3] \times [1,4]$ in the xy -plane.

$$\iint_R 4 \, dA, [0, 3] \times [1, 4]$$

Answer: _____

15. Find the area of the rectangle $R = [1,5] \times [1,6]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 5] \times [1, 6]$$

Answer: _____

16. Find the volume of the solid under $z = 8$ and above the rectangle $[1,4] \times [0,5]$ in the xy -plane.

$$\iint_R 8 \, dA, [1, 4] \times [0, 5]$$

Answer: _____

17. Find the area of the rectangle $R = [1,6] \times [0,6]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 6] \times [0, 6]$$

Answer: _____

18. Find the volume of the solid under $z = 9$ and above the rectangle $[0,3] \times [1,5]$ in the xy -plane.

$$\iint_R 9 \, dA, [0, 3] \times [1, 5]$$

Answer: _____

19. Find the area of the rectangle $R = [0,4] \times [0,7]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [0, 4] \times [0, 7]$$

Answer: _____

20. Find the volume of the solid under $z = 7$ and above the rectangle $[0,4] \times [0,3]$ in the xy -plane.

$$\iint_R 7 \, dA, [0, 4] \times [0, 3]$$

Answer: _____

Average value



21. A uniform temperature field has $T = 3$ degrees over rectangle $[1,3] \times [1,4]$. Use a double integral to confirm the total heat content is $3 \cdot \text{Area}$.

$$\iint_R 3 \, dA, [1, 3] \times [1, 4]$$

Answer: _____

22. A uniform temperature field has $T = 4$ degrees over rectangle $[1,2] \times [0,3]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [1, 2] \times [0, 3]$$

Answer: _____

23. A uniform temperature field has $T = 4$ degrees over rectangle $[1,5] \times [0,4]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [1, 5] \times [0, 4]$$

Answer: _____

24. A uniform temperature field has $T = 7$ degrees over rectangle $[0,5] \times [1,5]$. Use a double integral to confirm the total heat content is $7 \cdot \text{Area}$.

$$\iint_R 7 \, dA, [0, 5] \times [1, 5]$$

Answer: _____

25. A uniform temperature field has $T = 4$ degrees over rectangle $[0,3] \times [1,2]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [0, 3] \times [1, 2]$$

Answer: _____

26. A uniform temperature field has $T = 12$ degrees over rectangle $[0,3] \times [0,2]$. Use a double integral to confirm the total heat content is $12 \cdot \text{Area}$.

$$\iint_R 12 \, dA, [0, 3] \times [0, 2]$$

Answer: _____

27. A uniform temperature field has $T = 10$ degrees over rectangle $[0,3] \times [1,4]$. Use a double integral to confirm the total heat content is $10 \cdot \text{Area}$.

$$\iint_R 10 \, dA, [0, 3] \times [1, 4]$$

Answer: _____



28. A uniform temperature field has $T = 10$ degrees over rectangle $[0,5] \times [0,3]$. Use a double integral to confirm the total heat content is $10 \cdot \text{Area}$.

$$\iint_R 10 \, dA, [0, 5] \times [0, 3]$$

Answer: _____

29. A uniform temperature field has $T = 4$ degrees over rectangle $[0,5] \times [1,5]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [0, 5] \times [1, 5]$$

Answer: _____

30. A uniform temperature field has $T = 4$ degrees over rectangle $[1,5] \times [1,3]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [1, 5] \times [1, 3]$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Area and volume, Average value. All answers verified by independent computation.

Solutions



Area and volume

1. Find the area of the rectangle $R = [0,5] \times [0,4]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [0, 5] \times [0, 4]$$

→ Area = integral integral $1 \, dA = (5-0) \cdot (4-0) = 5 \cdot 4 = 20$ square units.

Answer: $= 1 \cdot 5 \cdot 4 = 20$

2. Find the volume of the solid under $z = 8$ and above the rectangle $[0,3] \times [1,3]$ in the xy -plane.

$$\iint_R 8 \, dA, [0, 3] \times [1, 3]$$

→ Volume = integral integral $8 \, dA$ over the rectangle.

→ $= 8 \cdot (3-0) \cdot (3-1) = 8 \cdot 3 \cdot 2 = 48$ cubic units.

Answer: $= 8 \cdot 3 \cdot 2 = 48$

3. Find the area of the rectangle $R = [1,6] \times [1,3]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 6] \times [1, 3]$$

→ Area = integral integral $1 \, dA = (6-1) \cdot (3-1) = 5 \cdot 2 = 10$ square units.

Answer: $= 1 \cdot 5 \cdot 2 = 10$

4. Find the volume of the solid under $z = 7$ and above the rectangle $[0,5] \times [1,4]$ in the xy -plane.

$$\iint_R 7 \, dA, [0, 5] \times [1, 4]$$

→ Volume = integral integral $7 \, dA$ over the rectangle.

→ $= 7 \cdot (5-0) \cdot (4-1) = 7 \cdot 5 \cdot 3 = 105$ cubic units.

Answer: $= 7 \cdot 5 \cdot 3 = 105$

5. Find the area of the rectangle $R = [0,7] \times [2,7]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [0, 7] \times [2, 7]$$

→ Area = integral integral $1 \, dA = (7-0) \cdot (7-2) = 7 \cdot 5 = 35$ square units.

Answer: $= 1 \cdot 7 \cdot 5 = 35$

6. Find the volume of the solid under $z = 9$ and above the rectangle $[1,4] \times [0,5]$ in the xy -plane.

$$\iint_R 9 \, dA, [1, 4] \times [0, 5]$$

→ Volume = integral integral $9 \, dA$ over the rectangle.

→ $= 9 \cdot (4-1) \cdot (5-0) = 9 \cdot 3 \cdot 5 = 135$ cubic units.

Answer: $= 9 \cdot 3 \cdot 5 = 135$



7. Find the area of the rectangle $R = [2, 7] \times [0, 4]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [2, 7] \times [0, 4]$$

→ Area = integral integral 1 dA = $(7-2) \cdot (4-0) = 5 \cdot 4 = 20$ square units.

Answer: $= 1 \cdot 5 \cdot 4 = 20$

8. Find the volume of the solid under $z = 5$ and above the rectangle $[1, 5] \times [1, 5]$ in the xy -plane.

$$\iint_R 5 \, dA, [1, 5] \times [1, 5]$$

→ Volume = integral integral 5 dA over the rectangle.

→ $= 5 \cdot (5-1) \cdot (5-1) = 5 \cdot 4 \cdot 4 = 80$ cubic units.

Answer: $= 5 \cdot 4 \cdot 4 = 80$

9. Find the area of the rectangle $R = [1, 7] \times [1, 6]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 7] \times [1, 6]$$

→ Area = integral integral 1 dA = $(7-1) \cdot (6-1) = 6 \cdot 5 = 30$ square units.

Answer: $= 1 \cdot 6 \cdot 5 = 30$

10. Find the volume of the solid under $z = 2$ and above the rectangle $[1, 4] \times [1, 3]$ in the xy -plane.

$$\iint_R 2 \, dA, [1, 4] \times [1, 3]$$

→ Volume = integral integral 2 dA over the rectangle.

→ $= 2 \cdot (4-1) \cdot (3-1) = 2 \cdot 3 \cdot 2 = 12$ cubic units.

Answer: $= 2 \cdot 3 \cdot 2 = 12$

11. Find the area of the rectangle $R = [1, 7] \times [2, 3]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 7] \times [2, 3]$$

→ Area = integral integral 1 dA = $(7-1) \cdot (3-2) = 6 \cdot 1 = 6$ square units.

Answer: $= 1 \cdot 6 \cdot 1 = 6$

12. Find the volume of the solid under $z = 5$ and above the rectangle $[0, 5] \times [0, 3]$ in the xy -plane.

$$\iint_R 5 \, dA, [0, 5] \times [0, 3]$$

→ Volume = integral integral 5 dA over the rectangle.

→ $= 5 \cdot (5-0) \cdot (3-0) = 5 \cdot 5 \cdot 3 = 75$ cubic units.

Answer: $= 5 \cdot 5 \cdot 3 = 75$

13. Find the area of the rectangle $R = [1, 7] \times [0, 5]$ using a double integral of $f(x, y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 7] \times [0, 5]$$

→ Area = integral integral 1 dA = $(7-1) \cdot (5-0) = 6 \cdot 5 = 30$ square units.

Answer: $= 1 \cdot 6 \cdot 5 = 30$



14. Find the volume of the solid under $z = 4$ and above the rectangle $[0,3] \times [1,4]$ in the xy -plane.

$$\iint_R 4 \, dA, [0, 3] \times [1, 4]$$

→ Volume = integral integral 4 dA over the rectangle.

$$\rightarrow = 4 \cdot (3-0) \cdot (4-1) = 4 \cdot 3 \cdot 3 = 36 \text{ cubic units.}$$

Answer: $= 4 \cdot 3 \cdot 3 = 36$

15. Find the area of the rectangle $R = [1,5] \times [1,6]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 5] \times [1, 6]$$

→ Area = integral integral 1 $dA = (5-1) \cdot (6-1) = 4 \cdot 5 = 20$ square units.

Answer: $= 1 \cdot 4 \cdot 5 = 20$

16. Find the volume of the solid under $z = 8$ and above the rectangle $[1,4] \times [0,5]$ in the xy -plane.

$$\iint_R 8 \, dA, [1, 4] \times [0, 5]$$

→ Volume = integral integral 8 dA over the rectangle.

$$\rightarrow = 8 \cdot (4-1) \cdot (5-0) = 8 \cdot 3 \cdot 5 = 120 \text{ cubic units.}$$

Answer: $= 8 \cdot 3 \cdot 5 = 120$

17. Find the area of the rectangle $R = [1,6] \times [0,6]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [1, 6] \times [0, 6]$$

→ Area = integral integral 1 $dA = (6-1) \cdot (6-0) = 5 \cdot 6 = 30$ square units.

Answer: $= 1 \cdot 5 \cdot 6 = 30$

18. Find the volume of the solid under $z = 9$ and above the rectangle $[0,3] \times [1,5]$ in the xy -plane.

$$\iint_R 9 \, dA, [0, 3] \times [1, 5]$$

→ Volume = integral integral 9 dA over the rectangle.

$$\rightarrow = 9 \cdot (3-0) \cdot (5-1) = 9 \cdot 3 \cdot 4 = 108 \text{ cubic units.}$$

Answer: $= 9 \cdot 3 \cdot 4 = 108$

19. Find the area of the rectangle $R = [0,4] \times [0,7]$ using a double integral of $f(x,y) = 1$ over R .

$$\iint_R 1 \, dA, [0, 4] \times [0, 7]$$

→ Area = integral integral 1 $dA = (4-0) \cdot (7-0) = 4 \cdot 7 = 28$ square units.

Answer: $= 1 \cdot 4 \cdot 7 = 28$



20. Find the volume of the solid under $z = 7$ and above the rectangle $[0,4] \times [0,3]$ in the xy -plane.

$$\iint_R 7 \, dA, [0, 4] \times [0, 3]$$

→ Volume = integral integral $7 \, dA$ over the rectangle.

→ $= 7 \cdot (4-0) \cdot (3-0) = 7 \cdot 4 \cdot 3 = 84$ cubic units.

Answer: $= 7 \cdot 4 \cdot 3 = 84$



Average value

21. A uniform temperature field has $T = 3$ degrees over rectangle $[1,3] \times [1,4]$. Use a double integral to confirm the total heat content is $3 \cdot \text{Area}$.

$$\iint_R 3 \, dA, [1, 3] \times [1, 4]$$

$$\rightarrow \text{Total} = \text{integral integral } 3 \, dA = 3 \cdot 2 \cdot 3 = 18.$$

Answer: $= 3 \cdot 2 \cdot 3 = 18$

22. A uniform temperature field has $T = 4$ degrees over rectangle $[1,2] \times [0,3]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [1, 2] \times [0, 3]$$

$$\rightarrow \text{Total} = \text{integral integral } 4 \, dA = 4 \cdot 1 \cdot 3 = 12.$$

Answer: $= 4 \cdot 1 \cdot 3 = 12$

23. A uniform temperature field has $T = 4$ degrees over rectangle $[1,5] \times [0,4]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [1, 5] \times [0, 4]$$

$$\rightarrow \text{Total} = \text{integral integral } 4 \, dA = 4 \cdot 4 \cdot 4 = 64.$$

Answer: $= 4 \cdot 4 \cdot 4 = 64$

24. A uniform temperature field has $T = 7$ degrees over rectangle $[0,5] \times [1,5]$. Use a double integral to confirm the total heat content is $7 \cdot \text{Area}$.

$$\iint_R 7 \, dA, [0, 5] \times [1, 5]$$

$$\rightarrow \text{Total} = \text{integral integral } 7 \, dA = 7 \cdot 5 \cdot 4 = 140.$$

Answer: $= 7 \cdot 5 \cdot 4 = 140$

25. A uniform temperature field has $T = 4$ degrees over rectangle $[0,3] \times [1,2]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [0, 3] \times [1, 2]$$

$$\rightarrow \text{Total} = \text{integral integral } 4 \, dA = 4 \cdot 3 \cdot 1 = 12.$$

Answer: $= 4 \cdot 3 \cdot 1 = 12$

26. A uniform temperature field has $T = 12$ degrees over rectangle $[0,3] \times [0,2]$. Use a double integral to confirm the total heat content is $12 \cdot \text{Area}$.

$$\iint_R 12 \, dA, [0, 3] \times [0, 2]$$

$$\rightarrow \text{Total} = \text{integral integral } 12 \, dA = 12 \cdot 3 \cdot 2 = 72.$$

Answer: $= 12 \cdot 3 \cdot 2 = 72$



27. A uniform temperature field has $T = 10$ degrees over rectangle $[0,3] \times [1,4]$. Use a double integral to confirm the total heat content is $10 \cdot \text{Area}$.

$$\iint_R 10 \, dA, [0, 3] \times [1, 4]$$

$$\rightarrow \text{Total} = \text{integral integral } 10 \, dA = 10 \cdot 3 \cdot 3 = 90.$$

Answer: $= 10 \cdot 3 \cdot 3 = 90$

28. A uniform temperature field has $T = 10$ degrees over rectangle $[0,5] \times [0,3]$. Use a double integral to confirm the total heat content is $10 \cdot \text{Area}$.

$$\iint_R 10 \, dA, [0, 5] \times [0, 3]$$

$$\rightarrow \text{Total} = \text{integral integral } 10 \, dA = 10 \cdot 5 \cdot 3 = 150.$$

Answer: $= 10 \cdot 5 \cdot 3 = 150$

29. A uniform temperature field has $T = 4$ degrees over rectangle $[0,5] \times [1,5]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [0, 5] \times [1, 5]$$

$$\rightarrow \text{Total} = \text{integral integral } 4 \, dA = 4 \cdot 5 \cdot 4 = 80.$$

Answer: $= 4 \cdot 5 \cdot 4 = 80$

30. A uniform temperature field has $T = 4$ degrees over rectangle $[1,5] \times [1,3]$. Use a double integral to confirm the total heat content is $4 \cdot \text{Area}$.

$$\iint_R 4 \, dA, [1, 5] \times [1, 3]$$

$$\rightarrow \text{Total} = \text{integral integral } 4 \, dA = 4 \cdot 4 \cdot 2 = 32.$$

Answer: $= 4 \cdot 4 \cdot 2 = 32$

